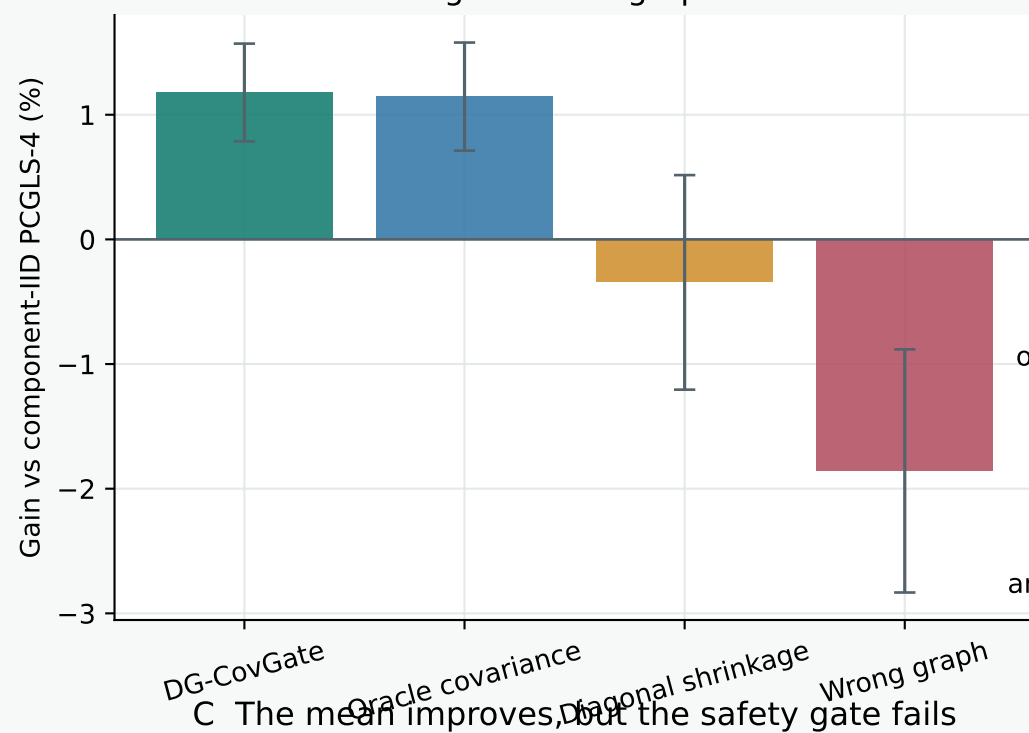
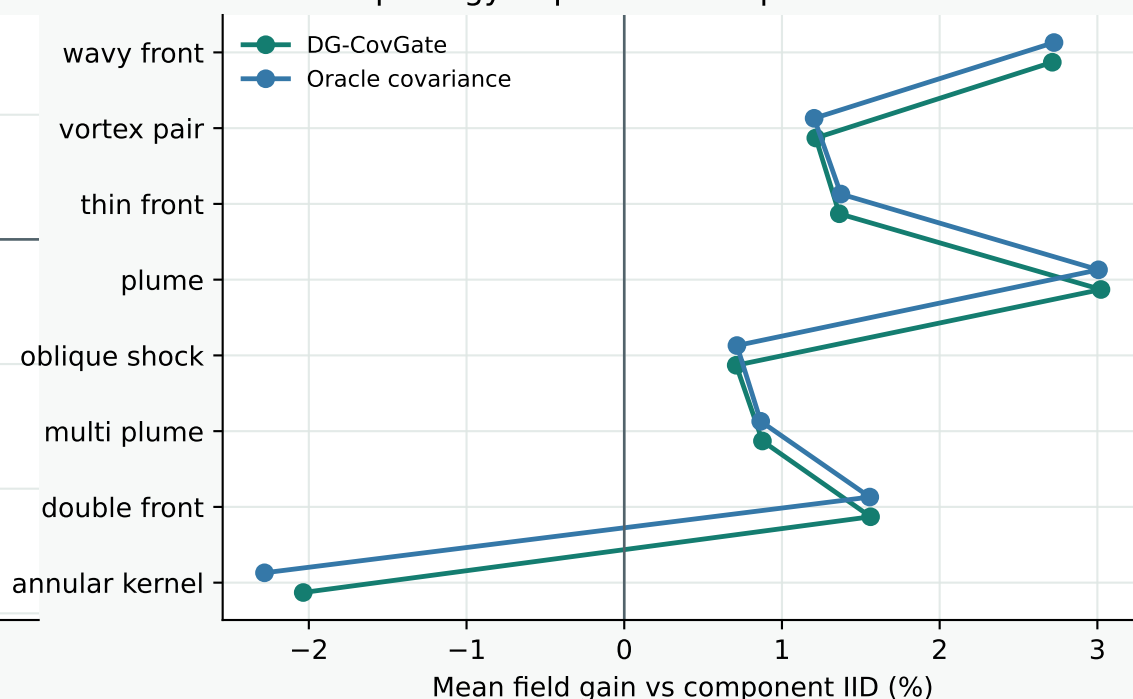


DG-WPCGLS: correct covariance helps on average, but fails the tail-risk gate

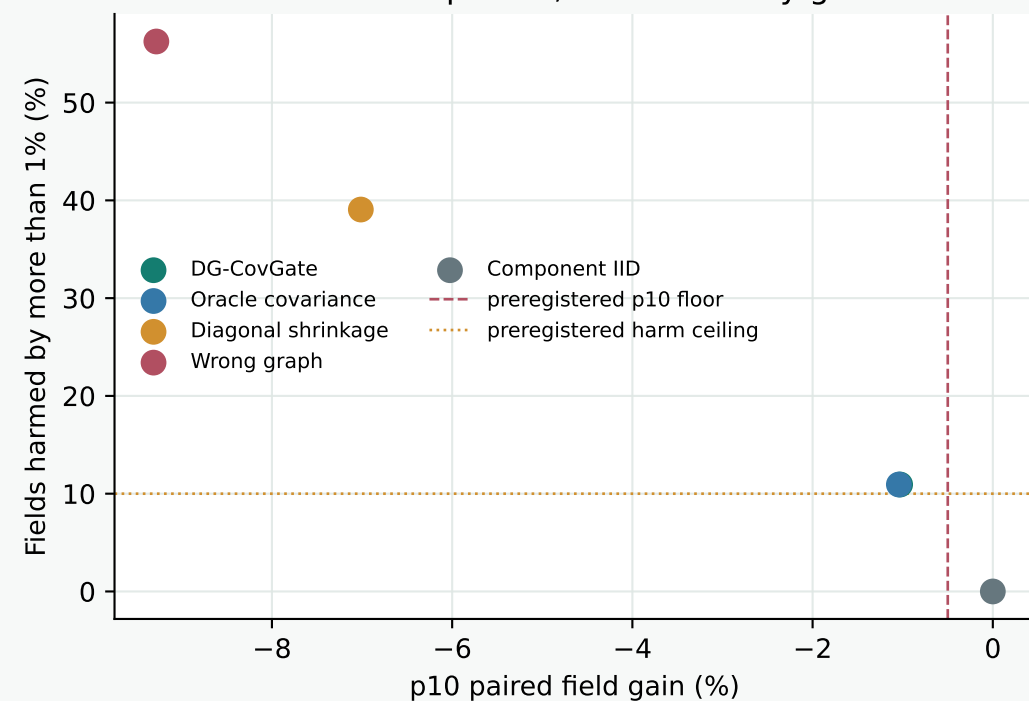
A Mean 3-D field gain under graph-correlated noise



B Morphology dependence exposes the unsafe tail



C The mean improves, but the safety gate fails



D Replicate-clustered gain across 16 fresh seeds

